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DSA0105-Object Oriented Programming with C++

CO2-AT1 Scenario Based Question

Question-1:

Define a class named movieMagic with the following description: (20)
Instance variables/data members:

int year – to store the year of release of a movie

String title – to store the title of the movie.

float rating – to store the popularity rating of the movie.

(minimum rating = 0.0 and maximum rating = 5.0)

Member Methods:

(i) void accept() To input and store year, title and rating.

(ii) void display() To display the title of a movie and a message based on the rating as per the table below.

RATING	MESSAGE TO BE DISPLAYED
0.0 to 2.0	Flop
2.1 to 3.4	Semi-hit
3.5 to 4.5	Hit
4.6 to 5.0	Super Hit

Write a main method to create an object of the class and call the above member methods.

PROGRAM

```
#include <iostream>
using namespace std;

class movieMagic
{
    int year = 2025;
    string title = "Avatar";
    float rating = 4.8;

public:
    void accept()
    {
        // Values already assigned
    }

    void display()
    {
        cout << "Movie Title: " << title << endl;

        if (rating <= 2.0)
            cout << "Flop";
        else if (rating <= 3.4)
            cout << "Semi-hit";
        else if (rating <= 4.5)
            cout << "Hit";
        else
            cout << "Super Hit";
```

```
    }
};

int main()
{
    movieMagic obj;
    obj.accept();
    obj.display();

    return 0;
}
```

OUTPUT:

```
Output

Enter year of release: 2026
Enter movie title: SIRAI
Enter rating: 4

Movie Title: SIRAI
Message: Hit
```

Question-2:

Define a class named BookFair with the following description: (20)
Instance variables/Data members:

String Bname – stores the name of the book.

double price – stores the price of the book.

Member Methods:

(i) void Input() – To input and store the name and the price of the book.

(ii) void calculate() – To calculate the price after discount. Discount is calculated

based on the following criteria.

PRICE	DISCOUNT
Less than or equal to Rs 1000	2% of price
More than Rs 1000 and less than or equal to Rs 3000	10% of price
More than Rs 3000	15% of price

(iv)void display() – To display the name and price of the book after discount.

Write a main method to create an object of the class and call the above member methodS.

PROGRAM

```
#include <iostream>
using namespace std;

class BookFair
{
    string Bname = "C++ Programming";
    double price = 2500;

public:
    void Input()
    {
        // Values already assigned
    }

    void calculate()
    {
        if (price <= 1000)
            price = price - (price * 2 / 100);
        else if (price <= 3000)
            price = price - (price * 10 / 100);
```

```
        else
            price = price - (price * 15 / 100);
    }

    void display()
    {
        cout << "Book Name: " << Bname << endl;
        cout << "Price after discount: Rs. " << price;
    }
};

int main()
{
    BookFair obj;

    obj.Input();
    obj.calculate();
    obj.display();

    return 0;
}
```

OUTPUT:

```
Book Name: C++ Programming
Price after discount: Rs. 2250
```

Question-3:

Define a class Electric Bill with the following specifications:

Instance Variable/ data member:

String n – to store the name of the customer

int units – to store the number of units consumed

double bill – to store the amount to paid

Member methods:

Void accept() – to accept the name of the customer and number of units consumed

Void calculate() – to calculate the bill as per the following tariff :

Number of units Rate per unit

First 100 units Rs.2.00

Next 200 units Rs.3.00

Above 300 units Rs.5.00

A surcharge of 2.5% charged if the number of units consumed is above 300 units.

Void print() – To print the details as follows :

Name of the customer ………

Number of units consumed ……

Bill amount …….

Write a main method to create an object of the class and call the above member methods

PROGRAM

```
#include <iostream>
using namespace std;

class ElectricBill
{
    string n = "Deepthi";
```

```
    int units = 350;
    double bill;

public:
    void accept()
    {
        // Values already assigned
    }

    void calculate()
    {
        if (units <= 100)
            bill = units * 2.00;

        else if (units <= 300)
            bill = (100 * 2.00) + ((units - 100) * 3.00);

        else
        {
            bill = (100 * 2.00) + (200 * 3.00)
                 + ((units - 300) * 5.00);

            bill = bill + (bill * 2.5 / 100);
        }
    }

    void print()
    {
        cout << "Name of the customer: " << n << endl;
```

```
        cout << "Number of units consumed: " << units << endl;
        cout << "Bill amount: Rs. " << bill << endl;
    }
};

int main()
{
    ElectricBill obj;

    obj.accept();
    obj.calculate();
    obj.print();

    return 0;
}
```

OUTPUT:

```
Name of the customer: Deepthi
Number of units consumed: 350
Bill amount: Rs. 1076.25
```

Question-4:

Design a class RailwayTicket with the following description: (20)

Instance variables/data members:

String name: to store the name of the customer.

String coach: to store the type of coach customer wants to travel. long mobno: to

store customer's mobile number.

int amt: to store basic amount of ticket.

int totalamt: to store the amount to be paid after updating the original amount.

Methods:

void accept(): to take input for name, coach, mobile number and amount.

void update(): to update the amount as per the coach selected. Extra amount to be added in the amount as follows:

Type of coaches	Amount
First_AC	700
Second_AC	500
Third_AC	250
sleeper	None

void display(): To display all details of a customer such as name, coach, total amount and mobile number.

Write a main() method to create an object of the class and call the above methods.

PROGRAM

```
#include <iostream>
using namespace std;

class RailwayTicket
{
    string name = "Deepthi";
    string coach = "First_AC";
    long mobno = 9876543210;
    int amt = 1000;
    int totalamt;
```

```
public:
    void accept()
    {
        // Values already assigned
    }

    void update()
    {
        if (coach == "First_AC")
            totalamt = amt + 700;
        else if (coach == "Second_AC")
            totalamt = amt + 500;
        else if (coach == "Third_AC")
            totalamt = amt + 250;
        else
            totalamt = amt;
    }

    void display()
    {
        cout << "Name: " << name << endl;
        cout << "Coach: " << coach << endl;
        cout << "Mobile Number: " << mobno << endl;
        cout << "Total Amount: Rs. " << totalamt << endl;
    }
};

int main()
{
```

```
    RailwayTicket obj;


    obj.accept();
    obj.update();
    obj.display();


    return 0;
}
```

OUTPUT:

```
Name: SRI
Coach: First_AC
Mobile Number: 9876543210
Total Amount: Rs. 1700
```

Question-5:

Design a class named ShowRoom with the following description: (20)

Instance variables/data members:

String name: to store the name of the customer.

long mobno: to store the mobile number of the customer.

double cost: to store the cost of the items purchased.

double dis: to store the discount amount.

double amount: to store the amount to be paid after discount.

Member methods:

void input(): to input customer name, mobile number, cost.

void calculate(): to calculate discount on the cost of purchased items, based on the following criteria:

COST DISCOUNT (IN PERCENTAGE)

Less than or equal to Rs. 10000 5%

More than Rs. 10000 and less than or equal to Rs. 20000 10%

More than Rs. 20000 and less than or equal to Rs. 35000 15%

More than Rs. 35000 20%

void display(): to display customer name, mobile number, amount to be paid after discount.

Write a main() method to create an object of the class and call the above member methods.

PROGRAM

```
#include <iostream>
using namespace std;

class ShowRoom
{
    string name = "Deepthi";
    long mobno = 9876543210;
    double cost = 25000;
    double dis, amount;

public:
    void input()
    {
        // Values already assigned
    }

    void calculate()
    {
        if (cost <= 10000)
```

```
            dis = cost * 5 / 100;
        else if (cost <= 20000)
            dis = cost * 10 / 100;
        else if (cost <= 35000)
            dis = cost * 15 / 100;
        else
            dis = cost * 20 / 100;

        amount = cost - dis;
    }

    void display()
    {
        cout << "Customer Name: " << name << endl;
        cout << "Mobile Number: " << mobno << endl;
        cout << "Amount to be paid: Rs. " << amount << endl;
    }
};

int main()
{
    ShowRoom obj;

    obj.input();
    obj.calculate();
    obj.display();

    return 0;
}
```

OUTPUT:

```
Customer Name: SRI
Mobile Number: 9876543210
Amount to be paid: Rs. 21250
```